Simra Ali

Robotics · Autonomous Systems · Embedded Design

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#1 in MA · #11 in the world	3× 1st place	SAT 1560	MIT BWSI · Harvard SSP
FTC 3565, out of 7,800+ teams (2026)	Hackathons & CAD competitions, 2024–25	790 Math · 770 EBRW · Academic High Honors	Autonomous systems coursework

EDUCATION

Westford Academy · Westford, MA — **Class of 2027**

SAT: 1560 (790 Math, 770 EBRW) · Academic High Honors

AP coursework: Calculus BC, Physics C, Chemistry, Computer Science A, Biology, U.S. History, Psychology, Microeconomics, and English

Harvard Summer School · Cambridge, MA — **Summer 2025**

Robotics, Autonomous Vehicles, Drones, and Artificial Intelligence · **Grade: A** · 4 credits

MIT Beaver Works Summer Institute — **2024–2026**

Autonomous Air Vehicle Racing	on-site · summer 2026	Foundations of Entrepreneurship — Founder	instructor-led · fall 2026
Basics of ASICs	instructor-led · fall 2024	Microelectronics & Hardware Development	online · self-paced
Python	online · self-paced	Git/GitHub and UNIX	online · self-paced

ROBOTICS EXPERIENCE & LEADERSHIP

FIRST Robotics Competition · Stormgears 5422 (45 members) — **2023 – Present**

Co-Captain · CAD and Mechanical Design Lead · Drive Team Coach

- Led the design and assembly of the 2026 competition robot, which received the FIRST Quality Award for design intent, robustness, and fabrication quality.
- Designed serviceable mechanisms in SOLIDWORKS 3D CAD, including a parameterized chassis, intake, hopper, shooter, inverted electrical board, and sensor and camera mounts.
- Coordinated match strategy and driver communication as Drive Team Coach; previous role: Drive Team Technician (2025).
- Managed inventory, schedules, project plans, and task tracking through Slack and Trello as Team Co-Captain.
- Trained 20+ students in electrical systems and CAD during summer and fall sessions.
- Mentored the incoming Electrical Lead after serving in that role in 2025.
- Presented to team sponsors: IBM, Triton Systems, Performance Motion Devices, and SOLIDWORKS.
- **Competition results:** Advanced to New England District Championship every season; 2026 Minuteman Alliance 6 Captain and WPI Alliance 7 Captain; 2024 UNH Finalist and Alliance 1 Captain.

FIRST Tech Challenge · Ghost Robotics 3565 (10 members) — **2023 – Present**

Computer Vision Software Developer · Onshape CAD Designer

- Contribute computer vision software, Onshape mechanical designs, and competition strategy, applying experience from FRC.
- Trained a Limelight neural network vision pipeline using a Roboflow dataset of 12,000+ labeled game-piece images as an alternative to traditional color thresholding.
- Experimenting with game-piece detection to develop density-aware path planning for autonomous pollen collection.
- **2026 results:** Massachusetts State Champion and Winning Alliance Captain; FIRST World Championship Ross Division Winner; set FTC World Record for highest match score; ranked #1 in Massachusetts and #11 worldwide among 7,800+ teams.

AUTONOMOUS SYSTEMS EXPERIENCE

Autonomous Air Vehicle Racing · MIT BWSI AeroAstro — **Summer 2026 – Ongoing research**

Software Lead · Researcher

Four-week on-site program

- Programmed computer vision, state estimation, control, navigation, and flight-dynamics models.
- Developed autonomous racing software in simulation with predictive speed control, yaw-based cornering, line-loss recovery, and altitude adjustment for gate navigation.
- Deployed the software to onboard drone hardware and evaluated performance through physical flight testing.
- Pursuing independent research in socially aware drone path planning and discussed the field with an MIT PhD researcher.

ENGINEERING PROJECTS & COMPETITIONS

SOLIDWORKS Demonstration Robots · built for a team sponsor — **May–Sept. 2024; May 2026–present**

- Built FRC-style demonstration robots: a tennis-ball robot with autonomous vision, intake, and shooting mechanisms; and a football robot with an intake, shooter, and elevator/arm for disk placement.
- Beta-tested SOLIDWORKS tools for FIRST workflows and collaborated on sharing custom FIRST components through GitHub.

The Bat · GirlHacks at Microsoft NERD — **1st Place** — **June 2025**

- Designed, built, and demonstrated an assistive navigation device for blind and low-vision users during the hackathon, using ultrasonic obstacle sensing and buzzer feedback.
- Open-sourced 3D-printable CAD models and published step-by-step assembly and wiring instructions using commonly available components, enabling others to build the device.

HireAble · NetScout Tech for Good — **1st Place** — **March 2025**

- Developed a web platform connecting people with disabilities to inclusive employers, integrating AI, database, and web technologies.

Firefly · Girls Who Code — **July–Aug. 2025**

- Built an interactive JavaScript game with p5play and p5.js, using TensorFlow.js hand-gesture detection for player input.

VTXCADs CADathon · Phillips Exeter Academy Robotics — **1st Place** — **Dec. 2024**

- Collaborated with a partner to design a 3D landscape water fountain in SOLIDWORKS.

Flood Watch · Harvard Summer School — **Summer 2025**

- Co-led the “Solve It, Defend It!” innovation club and developed a flood-risk prediction concept using drone and autonomous-vehicle data.

OUTREACH & MENTORSHIP

Middlesex County Robotics Club · Boxboro, MA — **Aug. 2023 – Present**

Outreach Coordinator

- Organize Girl Scout STEM Fest, FIRST LEGO League training and scrimmages, Apple Blossom Parade demos, and school STEM events; tutor at robotics summer camps to encourage girls’ participation in technical roles.
- Led the upgrade of the SaturnReach outreach robot.

Women in Computer Science · Westford Academy — **Sept. 2024 – Present**

Co-Founder and Co-Lead

- Teach Python, mentor member projects, prepare students for hackathons, and share STEM opportunities.

ADDITIONAL TECHNICAL PROGRAMS

Women in Technology Program · BAE Systems — **Jan.–May 2026**

Received mentorship and hands-on training from BAE Systems engineers in electrical, RF, software, mechanical, signal processing, optical, manufacturing, and failure-analysis engineering.

Girls Who Code (GWC) · Summer Immersion and Pathways — **June 2024 – July 2025**

Programmed projects in cybersecurity, cryptography, data science, machine learning, game design, and web development; Harvard GWC member.

SELECTED HONORS

FIRST Leadership Award (Dean’s List) Semi-Finalist	2026	FRC: Quality, Team Spirit, Gracious Professionalism	2025–26
National Merit Commended Scholar	2026	FTC: Control, Inspire, Innovate, and Connect	2025–26
AP Scholar	2026	Academic High Honors, Westford Academy	2024–26
Diversity, Equity and Inclusion Award, Westford Public Schools	2023	BWSI “You Belong in STEM” Essay Contest Honorable Mention	2026
Honored Citizen Award, Westford Public Schools	2023	Toastmasters Youth Leadership Award & Medal	2021

TECHNICAL SKILLS

Programming	Python, Java, JavaScript, HTML/CSS, SQL, Git/GitHub, WPILib, TensorFlow, NumPy, p5.js
CAD / Fabrication	SOLIDWORKS, Onshape, Autodesk Tinkercad, 3D printing
Hardware	Arduino, wiring, soldering, crimping, electrical-board layout, sensors, troubleshooting
Robotics / AI	Computer vision, SLAM, motion planning, reinforcement learning, autonomous systems, state estimation